

CSD 598 - Winter 2026

An intro to programming and statistical linear model
diagnostics

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3 March 2026

Last time: Exploration and inference example

Any questions? Any thoughts?

Intro to programming

- ▶ Obviously there is a lot to dig into here.
- ▶ A sufficient understanding does not require you to know too many details.
- ▶ Time allowing, we'll look at different programming concepts as we work through some ways of diagnosing the behaviour or statistical linear models.
- ▶ First, though, we'll have a look at your questions and possibly use some of the following slides to answer them.
 - ▶ We may also wander into Python or R to try things out.

If you are feeling courageous install these

- ▶ RStudio <https://posit.co/downloads/>
- ▶ Python <https://www.python.org/downloads/>
- ▶ Depending on your operating system some form of app store maybe the best bet for installing anything.
- ▶ If you are super adventurous, install uv (<https://docs.astral.sh/uv/getting-started/installation/>) and use Python that way. (It is actually a really good idea.)

Basic building blocks I

```
1  # These are lines comments. They are meant for humans to read
2  # and are ignored by the computer.
3
4  # Operators: Do stuff.
5  1+1
6
7  # Variables: Make the results stick around.
8  a = 1+1
9  b = 3
10
11 # Combinations thereof: What most programming actually is.
12 c = b/a
13
14 # Functions and their siblings: Package recipes for later use.
15 square(2)
16
```

Basic building blocks II

- ▶ integer – aka whole number,
- ▶ floating point number – aka decimal number,
- ▶ character – a single character,
- ▶ string – a piece of text made of characters, very useful to us,
- ▶ lists which will usually contain one given type of variable, e.g. a list of numbers or a list of strings.
- ▶ and many more.

Basic tools of the trade

- ▶ There are many programming languages and they are suited for different tasks and also differ in terms such as readability.
- ▶ Most relevant to us are R, Python, Matlab, Praat scripts, LibreOffice calc/Excel macros.
- ▶ Code files are usually identified with a suffix that corresponds to the language being used: e.g. `.R` for R, `.py` for Python, `.praat` for Praat (although the last one is not standardised).
- ▶ Code can be written with essentially any text editor that will save plain text (no formatting frills attached).

Actual tools of the trade

- ▶ Writing code generally easier with an actual code editor or IDE (Integrated Development Environment):
 - ▶ RStudio
 - ▶ Jasp
 - ▶ VSCode (corporate and a bit scary) / VSCodium (probably doesn't snitch on you)
 - ▶ PyCharm (often costs money)
 - ▶ Matlab
- ▶ Version control in some way, preferably with git if you end up doing this for any length of time.
- ▶ Consider using an AI helper (like Cursor, Copilot Agent, or Cline), but be aware that the less common your task, the more likely they are to produce only garbage.
- ▶ Helpers optimised for actual code writing are probably better than general generative models.

Skills of the trade

- ▶ Programming style: Code is read more than written. Learn to write in a clear style.
- ▶ Knowledge of the tools: There will always be new ones. Dedicate some time to learning new stuff.
- ▶ General principles like modularity, writing code to be read, documenting what you do, ...

Model diagnosis

- ▶ Collinearity (linear correlation) of the independent/predicting variables is bad (mathematically).
 - ▶ It can usually be spotted fairly easily by calculating correlations among the predictors and/or making scatter plots like a pairs plot.
- ▶ Heteroscedasticity will mess more the efficiency of prediction than the accuracy of it.
- ▶ Outliers can mess prediction accuracy.
- ▶ Here a bit of data exploration and model diagnostic plots go a long way:
 - ▶ Neat explanation of diagnostic plots can be found here
<https://library.virginia.edu/data/articles/diagnostic-plots>
 - ▶ Collinearity can be seen in for example a pairs plot or a correlation matrix (not to be confused with e.g. "Correlation of Fixed Effects" in 'lmer' output).

Some resources

- ▶ RStudio <https://posit.co/downloads/>
- ▶ Jasp <https://jasp-stats.org/>
- ▶ Python learning materials
<https://wiki.python.org/moin/BeginnersGuide/NonProgrammers>
 - ▶ Especially the Interactive Courses are really good for learning programming in general.
- ▶ Praat scripting guide <https://praatscriptingtutorial.com/>